ModelIt!

Student Activity Pack

Electromagnetic Spectrum: Beyond Visible Light

The Invisible Waves That Run Your World

Grade 07L2 | NGSS MS-PS4-2, MS-PS4-3 | Structure and Function

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is light? Is it a particle, a wave, or something else?

2 Can you name any types of waves or radiation besides visible light?

3 Draw what you think happens when light hits a glass window vs. a brick wall.

4 How do you think your phone receives text messages without a wire connecting it?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Wave Frequency | Wave Amplitude | Wave Energy | Penetration Depth | Detection Sensitivity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Radio Frequency

Very low frequency, moderate amplitude

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Visible Light

Mid-range frequency, moderate amplitude

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: X-ray Frequency

Very high frequency, moderate amplitude

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Material Density — How tightly packed the atoms are in the material the wave encounters, affecting absorption and transmission

Signal Noise — Random interference that degrades the quality of transmitted information, affecting both analog and digital signals

Absorption Rate — How much of the wave's energy is converted to heat or other forms when passing through a material

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If visible light is just a tiny sliver of the electromagnetic spectrum, what else is out there that we can't see? Use evidence from your simulation to support your answers.

1. **Why is visible light the only part of the spectrum your eyes can detect? What evolutionary advantage might this provide?**

2. **How does your model explain why sunscreen protects against UV radiation but not visible light?**

3. **Why are digital signals more reliable than analog signals for transmitting information?**

4. **What would the world look like if your eyes could see infrared or ultraviolet instead of visible light?**

5. **How does the relationship between frequency and energy explain why gamma rays are dangerous but radio waves are not?**

8 STEM Challenge: Design an EM Spectrum Technology Showcase

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YOUR ENGINEERING MISSION

Create a showcase demonstrating how different parts of the electromagnetic spectrum are used in everyday technology, with evidence from your wave model explaining why each application works.

THE SCENARIO

A tech company wants to create an educational display showing customers how electromagnetic waves power their everyday devices: from WiFi to medical imaging to TV remotes. Your team will design the display using wave science to explain each technology.

Engineering Design Thinking

1. WHICH TECHNOLOGIES WILL YOU FEATURE AND WHAT EM SPECTRUM REGION DOES EACH USE?
2. HOW WILL YOU VISUALLY REPRESENT THE SPECTRUM AND EACH TECHNOLOGY'S PLACE ON IT?
3. WHAT MODEL DATA WILL YOU INCLUDE TO EXPLAIN WHY EACH APPLICATION USES ITS SPECIFIC FREQUENCY?
4. HOW WILL YOU EXPLAIN DIGITAL VS. ANALOG SIGNAL TRANSMISSION TO A NON-TECHNICAL AUDIENCE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Telecommunications Engineers design and maintain systems that transmit information using electromagnetic waves, including cell networks, satellite communications, and fiber optics, earning $80,000-$140,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In the electromagnetic spectrum model, increasing Wave Frequency causes Wave Energy to increase. A hospital uses X-rays (high frequency) to image bones inside the body. Based on the model, why do X-rays pass through soft tissue but are stopped by bone?

- **A.** X-rays are attracted to calcium in bones and repelled by soft tissue
- **B.** X-ray photons carry enough energy to penetrate low-density soft tissue but are absorbed by the high-density mineral structure of bone
- **C.** Bones reflect X-rays while soft tissue absorbs them completely
- **D.** X-rays only travel in straight lines and bones are in the way

2 A student wonders why WiFi signals (radio waves) pass through walls but visible light does not. Using the model's relationships between frequency, energy, and material interaction, which explanation is most accurate?

- **A.** Radio waves are smaller than visible light waves and can fit between wall molecules
- **B.** Radio waves have lower frequency and interact weakly with wall materials, passing through without significant absorption, while visible light's frequency matches the absorption properties of wall materials
- **C.** Walls have special radio-transparent coatings built into them
- **D.** Radio waves are stronger than visible light, so they can push through walls

3 The model shows that Wave Frequency and Wave Energy have a positive relationship. A student concludes that higher frequency waves are always more dangerous. Which evidence from the model challenges this oversimplification?

- **A.** All electromagnetic waves are equally dangerous regardless of frequency
- **B.** Danger depends on both energy per photon (frequency) AND intensity (amplitude). A weak X-ray beam can be less harmful than an extremely intense radio beam because total energy delivered matters, not just frequency
- **C.** Lower frequency waves are actually more dangerous because they penetrate deeper
- **D.** The model does not include any information about wave danger

4 Visible light represents less than 1% of the electromagnetic spectrum, yet human eyes evolved to detect this specific frequency range. Based on the model's concepts, which explanation best accounts for why this particular range?

- **A.** Visible light is the only type of electromagnetic wave that exists on Earth's surface
- **B.** The visible range matches the peak output of our Sun and passes efficiently through Earth's atmosphere, making it the most useful range for organisms to detect
- **C.** All organisms on Earth detect the same frequency range because DNA encodes for visible light detection
- **D.** Visible light is the safest frequency range, so evolution selected for eyes that avoid detecting dangerous wavelengths

5 **A telecommunications engineer must choose between analog and digital signals for a long-distance communication system. Based on the model's concepts about signal transmission, which advantage makes digital signals more reliable?**

- **A.** Digital signals travel faster than analog signals through the same medium
- **B.** Digital signals use higher frequencies than analog signals
- **C.** Digital signals encode information as discrete 1s and 0s that can be perfectly regenerated at relay points, while analog signals accumulate noise with distance and cannot be restored to their original form
- **D.** Digital signals are immune to all forms of electromagnetic interference